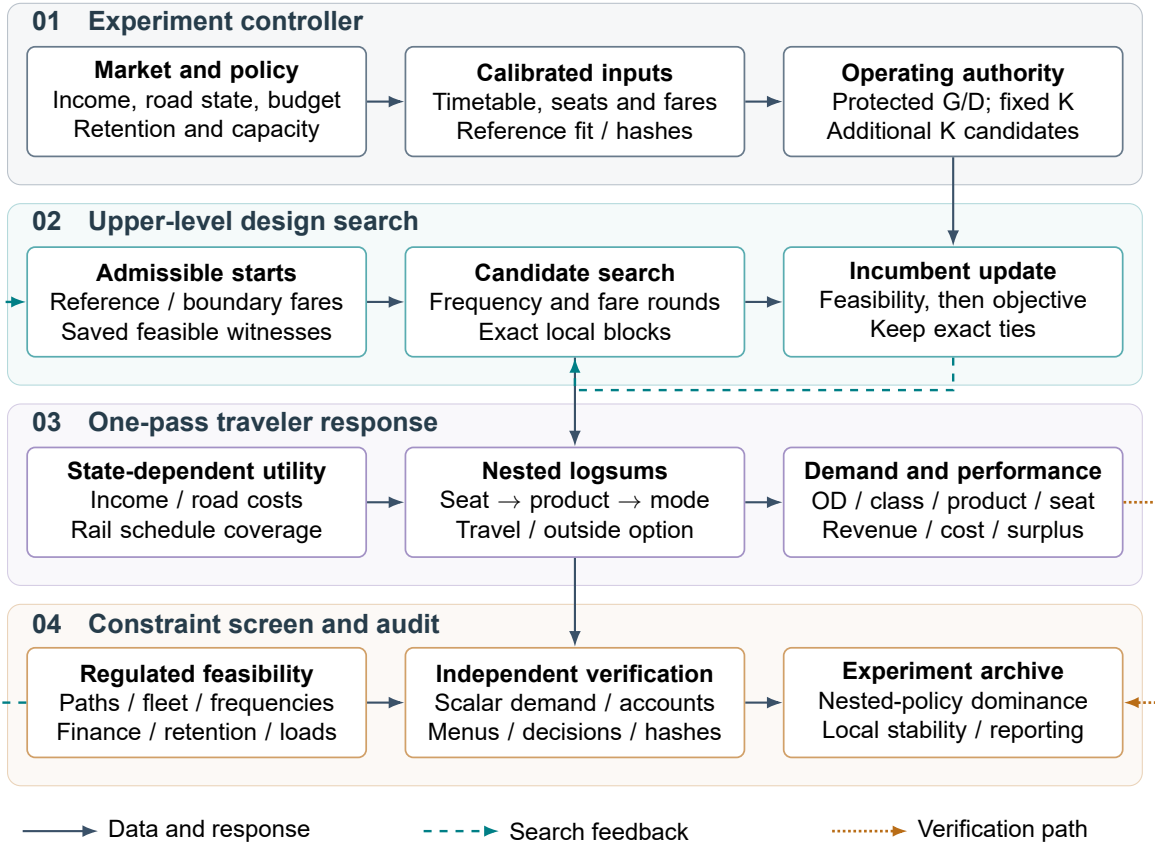
01 Experiment controller
Market and policy
Income, road state, budget
Retention and capacity
Calibrated inputs
Timetable, seats and fares
Reference fit / hashes
Operating authority
Protected G/D; fixed K
Additional K candidates
02 Upper-level design search
Admissible starts
Reference / boundary fares
Saved feasible witnesses
Candidate search
Frequency and fare rounds
Exact local blocks
Incumbent update
Feasibility, then objective
Keep exact ties
03 One-pass traveler response
State-dependent utility
Income / road costs
Rail schedule coverage
Nested logsums
Seat → product → mode
Travel / outside option
Demand and performance
OD / class / product / seat
Revenue / cost / surplus
04 Constraint screen and audit
Regulated feasibility
Paths / fleet / frequencies
Finance / retention / loads
Independent verification
Scalar demand / accounts
Menus / decisions / hashes
Experiment archive
Nested-policy dominance
Local stability / reporting
Data and response
Search feedback
Verification path